Bryan Wei Tern LIM

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EDUCATION

Massachusetts Institute of Technology ***2019 – Present***

Selected for the competitive MIT-Imperial College exchange program to complete final academic year (2019/2020) at MIT.

GPA: 5.0

Imperial College London ***2016 – Present***

4th Year MEng Mechanical Engineering student**,** 1st, 2nd and 3rd year grade – First Class

Methodist College Kuala Lumpur ***2014 – 2015***

4A*s for CIE A level examinations

AWARDS

Faculty of Engineering UROP Bursary Award – Imperial College London **2019**

Gamuda Scholarship ***2016 – Present***

Full scholarship for degree at Imperial College London – Only overseas scholar selected from over 1600 applicants

World Scientific Scholarship Award – Imperial College London ***2016 – Present***

Merit and Leadership Scholarship - Methodist College Kuala Lumpur ***2014 – 2015***

Full scholarship for A level course

RESEARCH EXPERIENCE

Undergraduate Researcher at the **Biomimetics Lab, MIT** ***2019 – Present***

Supervisor: Prof. Sangbae Kim and Dr. Donghyun Kim

- Researching the use of machine learning methods on legged robots for dynamic locomotion tasks.
- Investigating reinforcement learning methods on a robust and powerful quadruped robot (MIT Mini Cheetah) using deep neural networks, quality diversity optimization and evolutionary algorithms to robustly perform locomotion tasks.
- Implementing a guided policy search (imitation learning) for the Regularized Predictive Controller for quadrupedal robots.
- Designing an intelligent footstep planner for uneven terrain locomotion with large height variations to be integrated with existing control architecture consisting of Model Predictive Control and Whole-Body Control.

Undergraduate Researcher at **MCUBE Lab, MIT** ***2019 – Present***

Supervisor: Prof. Alberto Rodriguez and Maria Bauza

- Researching the topic "Tactile reasoning for intelligent dual arm manipulation". This project aims to make robots significantly more capable of robust and dexterous manipulation through high resolution sensors and learning algorithms.
- Investigating the use of various learning methods (CNN's and GAN's) for learning the local shape from tactile images for accurate localisation of the robotic arm for smaller objects.
- Implemented a perception system for pose estimation using RGBD cameras and high-resolution depth cameras (laser scanners). This involves image segmentation and ICP registration with the object model to obtain a real-time pose estimate.
- Improving the ability of the algorithms to generalize to unseen data by designing a platform that generates meaningful contact interactions and by introducing adversarial perturbations and other data augmentation techniques.
- Programmed the controls of a simple robotic system to collect contact data efficiently for training of the neural networks.
- Preparing a custom dataset to train a Grasp-Quality Convolutional Neural Networks (GQ-CNN) to propose grasps given depth images for industrial kitting tasks.
- Investigating observability affordances and metrics to complement stability metrics when proposing grasps.

Undergraduate Researcher at **Adaptive and Intelligent Robotics Lab (AIRL), Imperial College** ***2018 – 2019***

Supervisor: Dr. Antoine Cully

- Primary objective of the projects undertaken at AIRL is to use learning algorithms (deep reinforcement learning, quality diversity optimization, evolutionary algorithms and others) to improve the adaptivity, versatility and autonomy of physical robots.
- Implemented an updated version of the Intelligent Trial and Error (ITE) algorithm (Nature, 2015) on a hexapod as a demo for the lab. This was accomplished by generating a behavioural repertoire using the MAP-Elites algorithm and using Bayesian optimisation for adaptation.
- Implemented an updated version of the Reset-free Trial and Error (RTE) algorithm on a hexapod robot and a quadruped robot (open source quadruped DogBot) in simulation. This also included repertoire generation but used other methods such as Monte Carlo Tree Search (MCTS) and A* for path planning to navigate obstacles and reach targeted positions.
- Exhibitor for the lab at the AI Summit London 2019 and the Great Exhibition Road Festival 2019.

PUBLICATIONS

Conference Publications

2020 IROS "Robust Autonomous Navigation of a Small-Scale Quadruped Robot in Real-World Environments" to International Conference on Intelligent Robots and Systems (IROS) (*In Review*)

2020 RSS "Object Pose Estimation with Geometric Tactile Rendering and Tactile Image Matching" to Robotics: Science and Systems (RSS) (*In Review*)

2020 ICRA "Vision Aided Dynamic Exploration of Unstructured Terrain with a Small-Scale Quadruped Robot" to the International Conference on Robotics and Automation (ICRA).

PROJECTS

IDbot – Automated ID-Photo taking robot (3rd Year Design, Make and Test group project) ***2018 – 2019***

- Led a team of four engineers to design and prototype a robot that automates capturing ID headshots by removing by removing the need for verbal direction for head tilts and turns.
- Grade obtained: A
- A video demo of the project is available at: https://www.youtube.com/watch?v=nWTzkd-DQiQ\

Autonomous bomb disposal robot (3rd Year Embedded Systems and Microcontrollers project) ***2018***

- Designed, programmed (C) and tested a fully functioning mechatronic system, which autonomously navigates to an IR beacon, reads and displays a RFID code and returns to its original position.
- Grade obtained: A
- A video demo is of the project is available at: https://www.youtube.com/watch?v=bxo3q0mNtYE

WORK EXPERIENCE

Tunnel Engineering Intern at Gamuda Berhad Malaysia ***2018***

- Worked on a variety of the projects involving the latest tunnelling methods; includes preparing the Tunnel Boring Machine (TBM) for launch, plant setup and ground improvement works along the tunnel alignment.
- Implemented modifications to a TBM to accommodate new information on different ground conditions, by designing two new supply lines and selecting required mechanical and electrical components. I also designed and prototyped a mixing nozzle. This in-house modification was a first of its kind and resulted in cost savings of over 1 million MYR.
- Managed and coordinated work programs with sub-contractors, consultants and the client to keep project timeline.

Mechanical Engineering Intern at Gamuda Berhad Malaysia ***2017***

- Interned at the Design and Technical department of the Mass Rapid Transport 2 (MRT2) underground project. Worked on a variety of tasks involving mechanical and electrical systems of buildings and underground structures.
- Reviewed, analysed and corrected engineering drawings and flow calculations of cold-water, drainage and sewerage systems.

VOLUNTEERING AND OTHER WORK EXPERIENCE

Volunteer at New Hope Community Services, Singapore ***2017***

- Volunteered for two months at New Hope Community Services which was established to provide shelter to homeless Singaporeans and to empower them with opportunities - to alleviate and improve their situation.
- Organised a three-day adventure camp for underprivileged youth and children aged 7-21 years old from the shelter.
- Launched New Hope Club which intends to end the cycle of poverty by creating a safe space for homeless youth & children.

Tennis Player (Former Malaysian National Tennis Player) ***2010 – 2017***

- Selected by the Lawn Tennis Association of Malaysia at the age of 14 to join the Malaysian National Tennis Team. Chosen to be part of a team of five Malaysian players sent to Barcelona for a three-month training stint.
- Represented Malaysia at International Tennis Federation (ITF) Junior tournaments.
- Gold medallist at the National Junior Championships (MSSM) in the years 2011, 2012 and 2013.
- Represented the Imperial College Lawn Tennis 1st team at the British University and Collegiate Sports (BUCS) Games.

Volunteer at Khmer Village Homestay, Kampong Thom Province, Baray, Cambodia ***2016***

- Volunteered for 2 months in Khmer Village Homestay which was set up to alleviate rural poverty after the Khmer rouge through promoting local industry and community involvement projects with the local authorities.
- Communicated and liaised with representatives from Ngee Ann Polytechnic, Singapore to plan projects for a group of volunteers part of the Youth Expedition Program (YEP) under the Singapore National Youth Council (NYC).
- Worked part of a team to plan and coordinate community building projects such as painting, fence and well construction.
- Restructured the content of teaching material used for Wisdom Nest School in Baray district.

Tutor at VIP Education Group, Kuala Lumpur, Malaysia ***2015 – 2016***

- Worked part time as a chemistry, biology, and physics tutor preparing students for SPM, IGCSE and A level examinations.
- Developed marketing skills and business acumen by managing tuition centres and liaising with clients and parents.